VLSI System Design
(BECE303L)

Title: Phase Locked Loop (PLL)

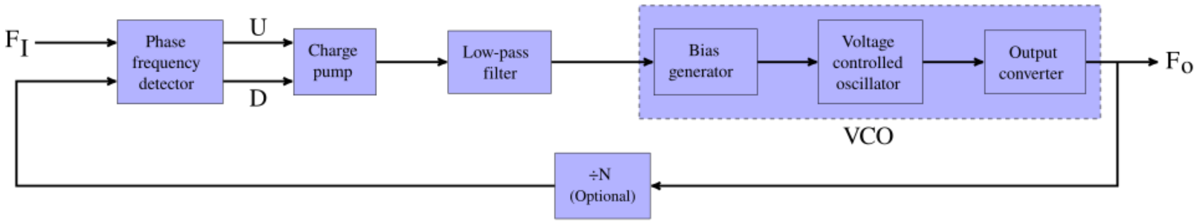
Tharun Ravuru-22BEC0895

Introduction:

A phase-locked loop or phase lock loop (PLL) is a control system that generates an output signal whose phase is fixed relative to the phase of an input signal. Keeping the input and output phase in lockstep also implies keeping the input and output frequencies the same, thus a phase-locked loop can also track an input frequency. Furthermore, by incorporating a frequency divider, a PLL can generate a stable frequency that is a multiple of the input frequency.

These properties are used for clock synchronization, demodulation, frequency synthesis, clock multipliers, and signal recovery from a noisy communication channel.

Block Diagram:



1. Phase Frequency Detector:

Compares the reference clock (Vref) with the feedback clock (Vdiv) from the divider.

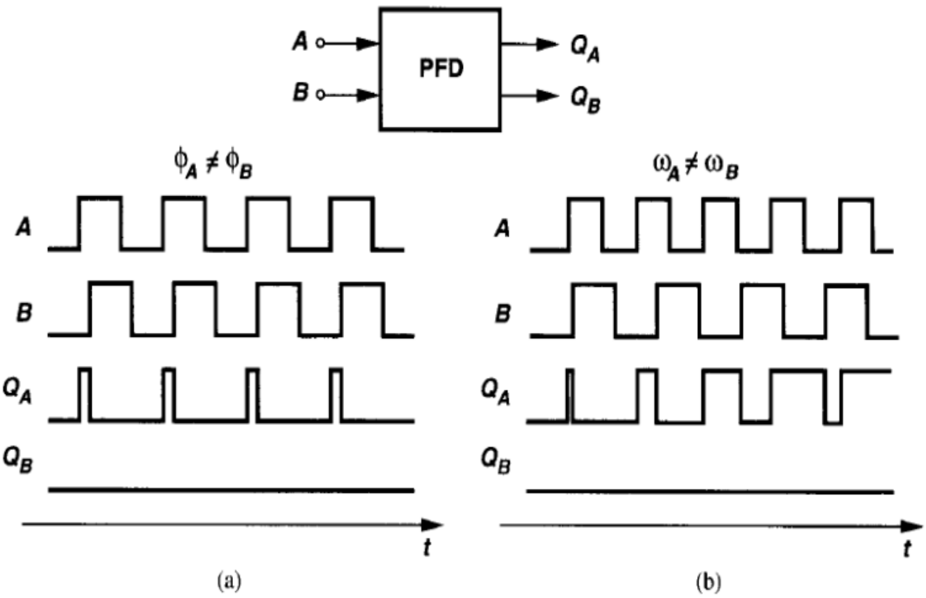
Produces two digital outputs:

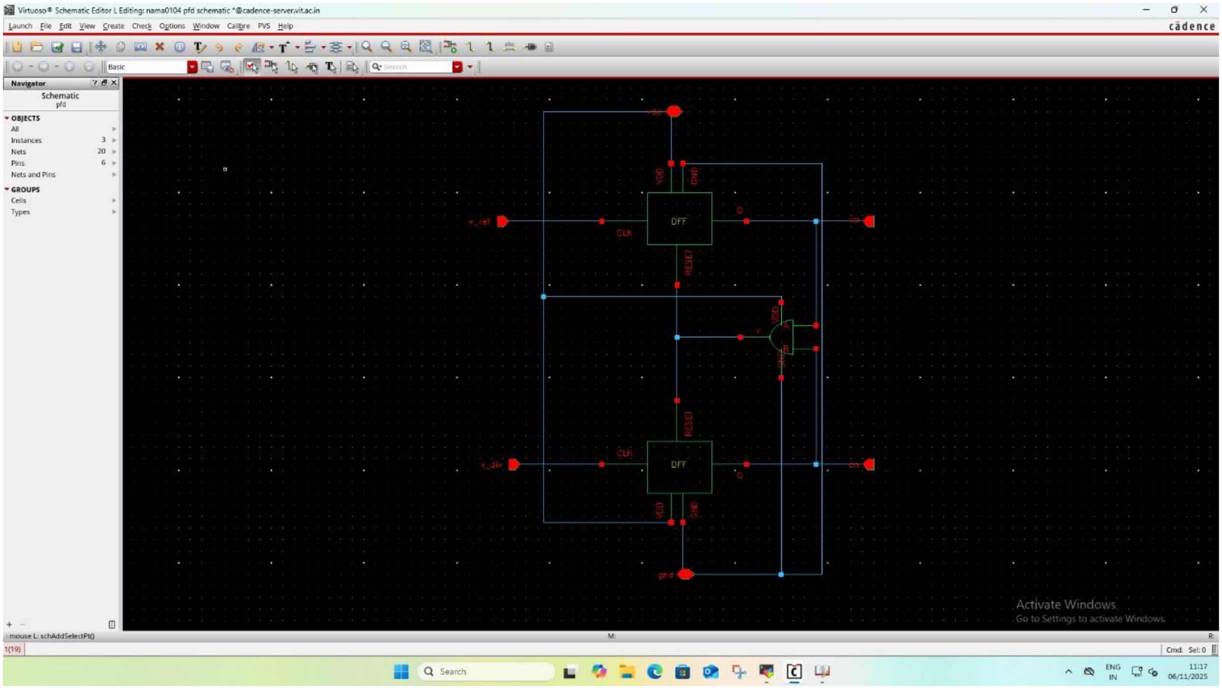
UP → When reference is leading (PLL frequency too low)

DN → When feedback is leading (PLL frequency too high)

Gives both phase and frequency difference.

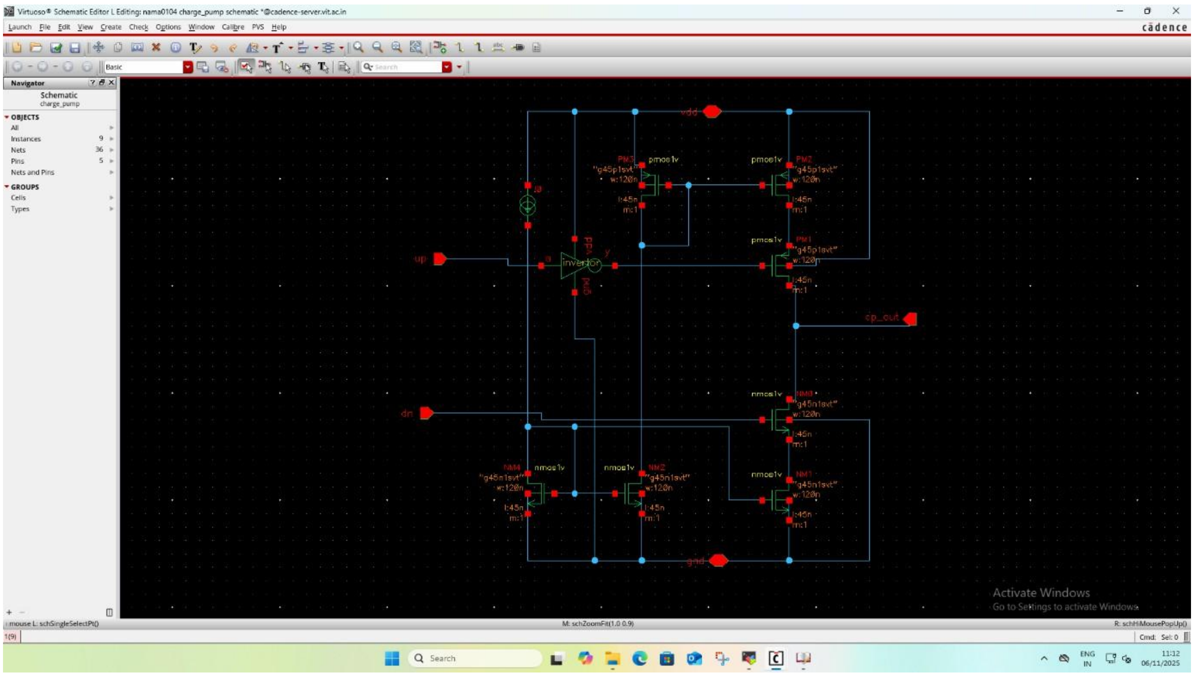
Output is used to control the charge pump.





2. Charge Pump:

Charge pumps are crucial components in the design of phase-locked loops (PLLs), a type of feedback control system used to synchronize an output signal with a reference signal.

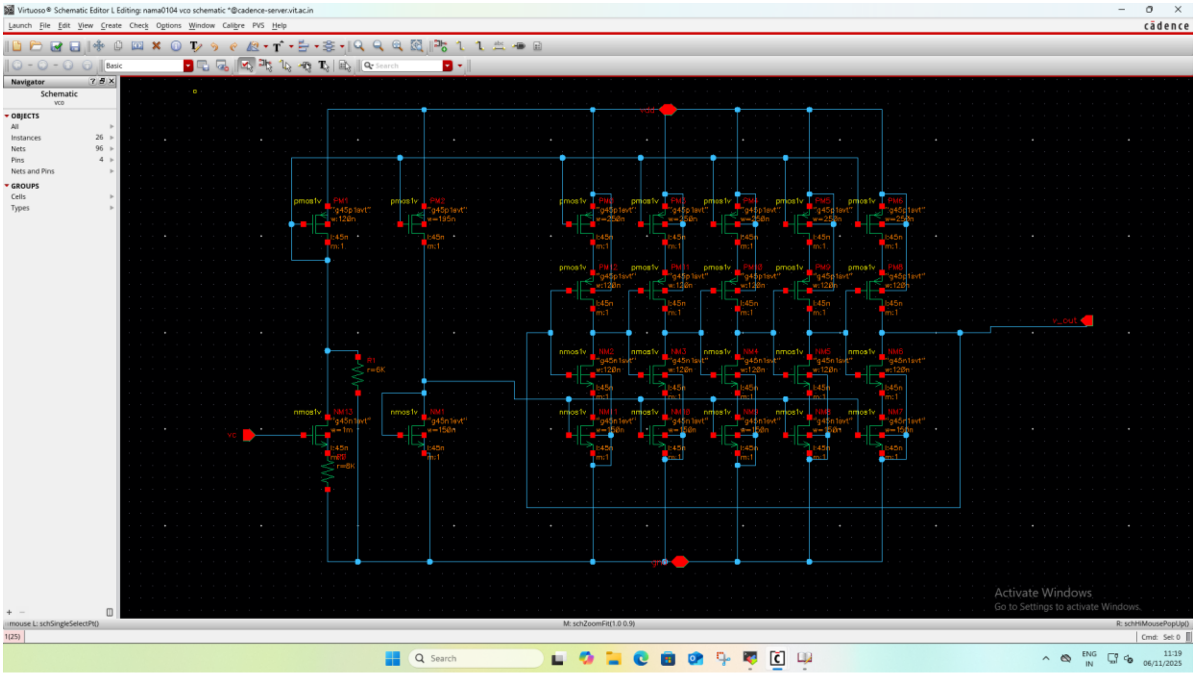


3. Voltage-Controlled Oscillator (VCO):

Converts the control voltage (**Vc**) into a clock signal.

Frequency increases when **Vc increases**, decreases when **Vc decreases**.

Output is the **PLL output frequency**.

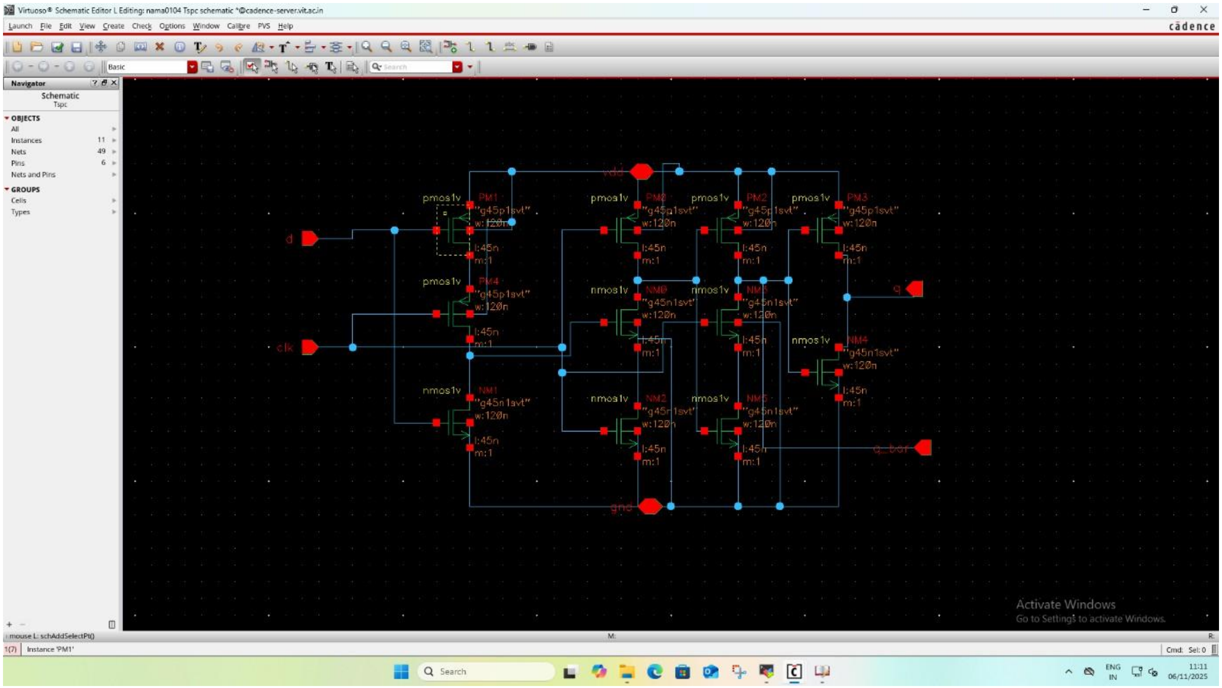


4. TSPC (True Single-Phase Clock) FF

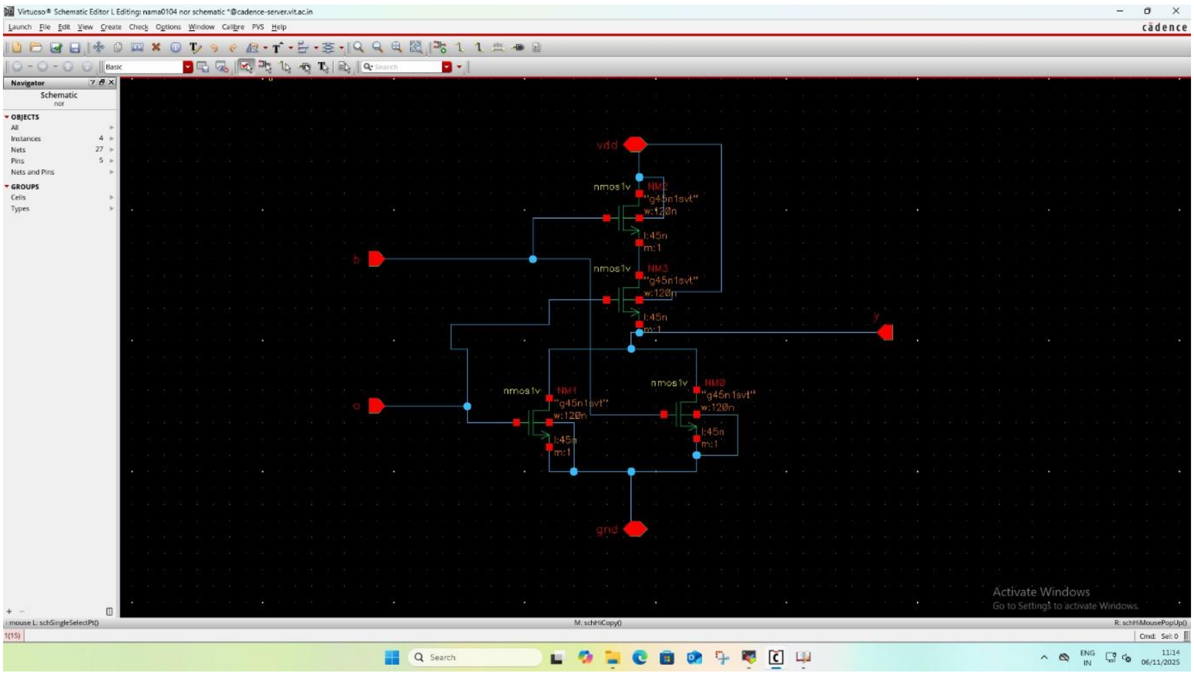
High-speed operation with a single clock.

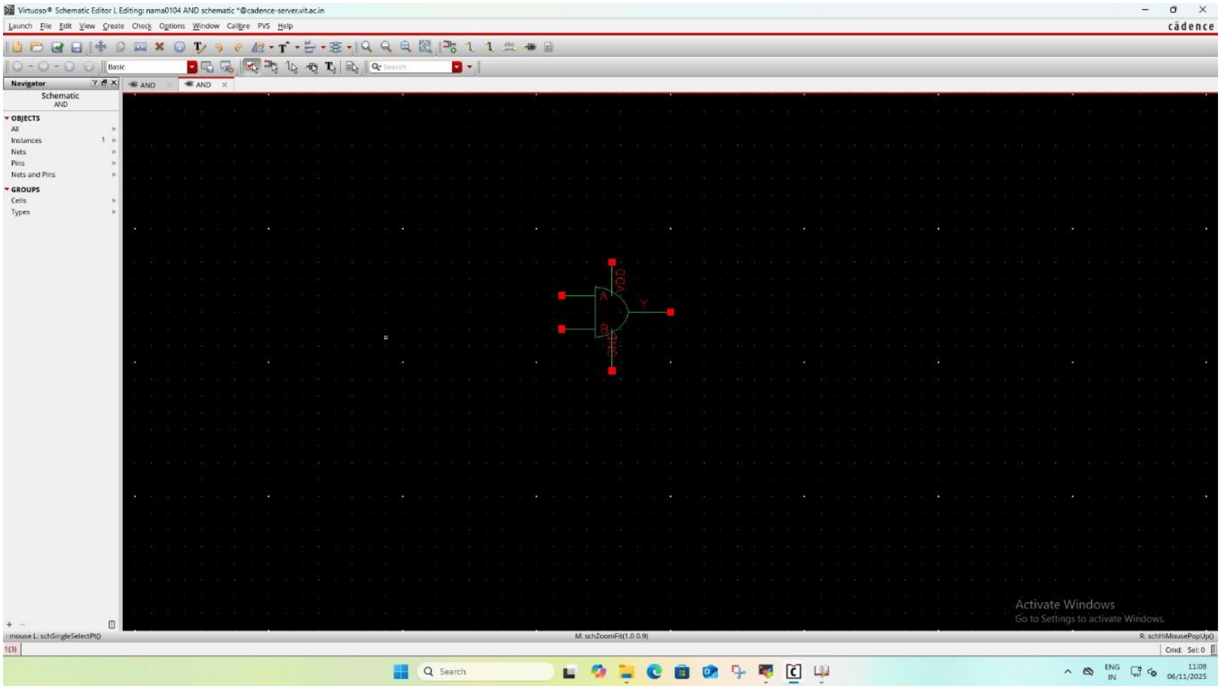
Consumes less power.

Good for high-frequency PLL dividers.



5. NOR, AND D FLIPFLOP:

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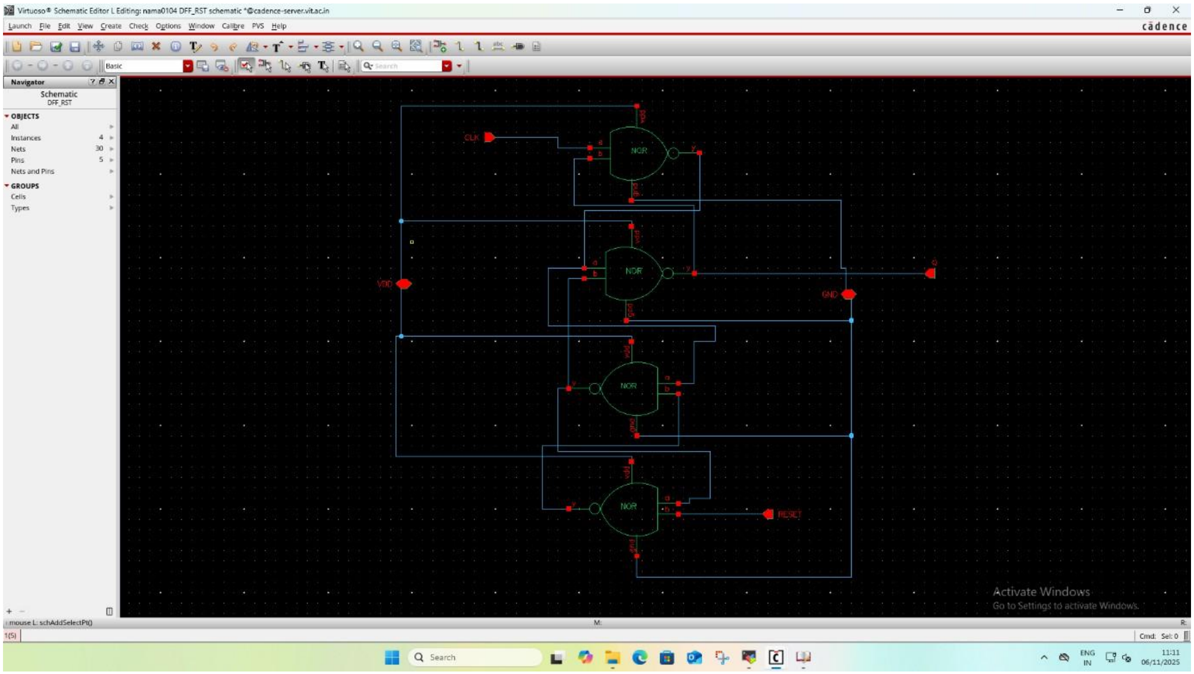
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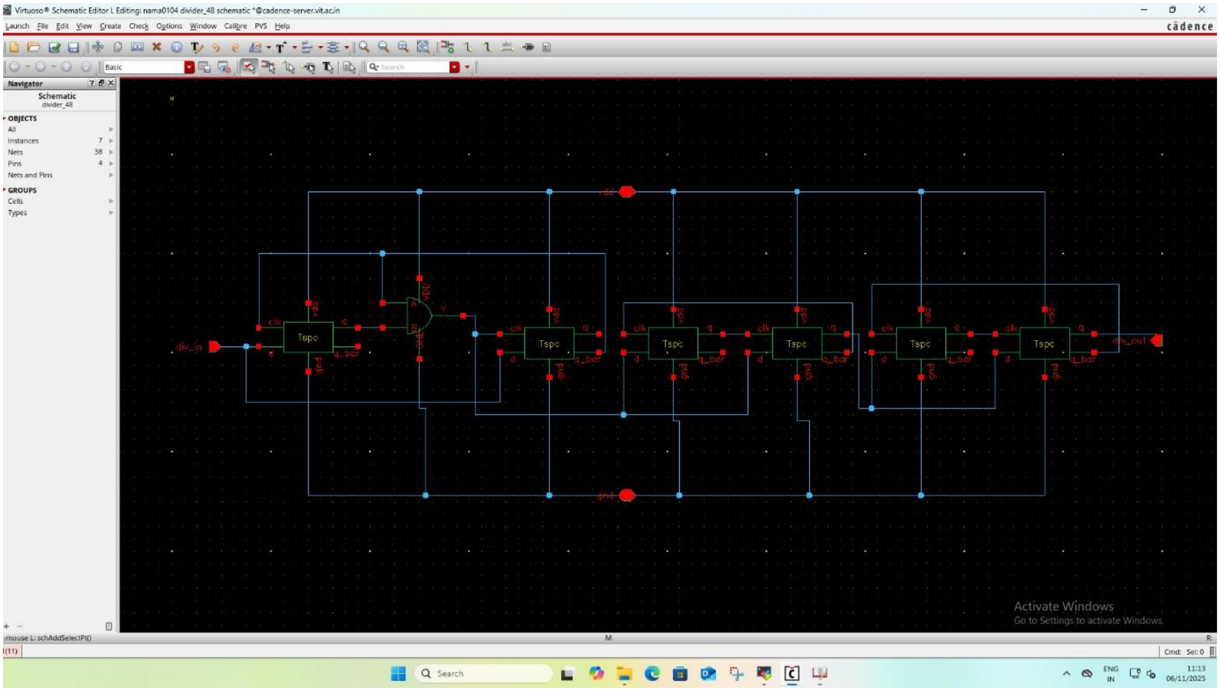
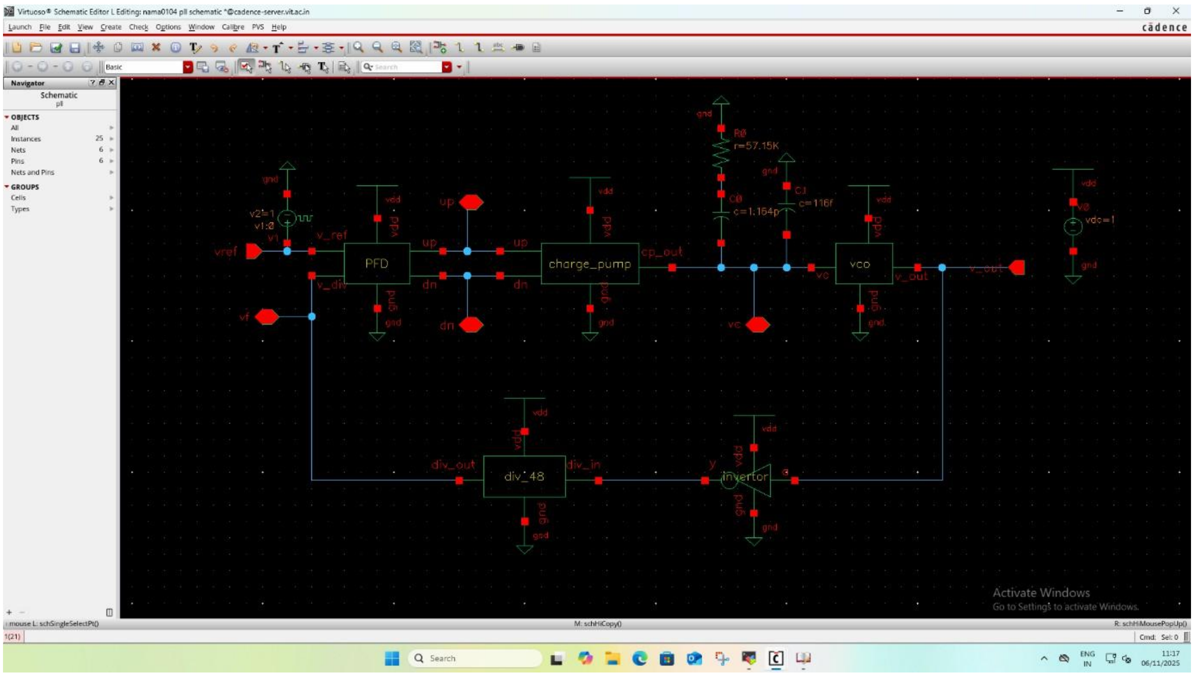
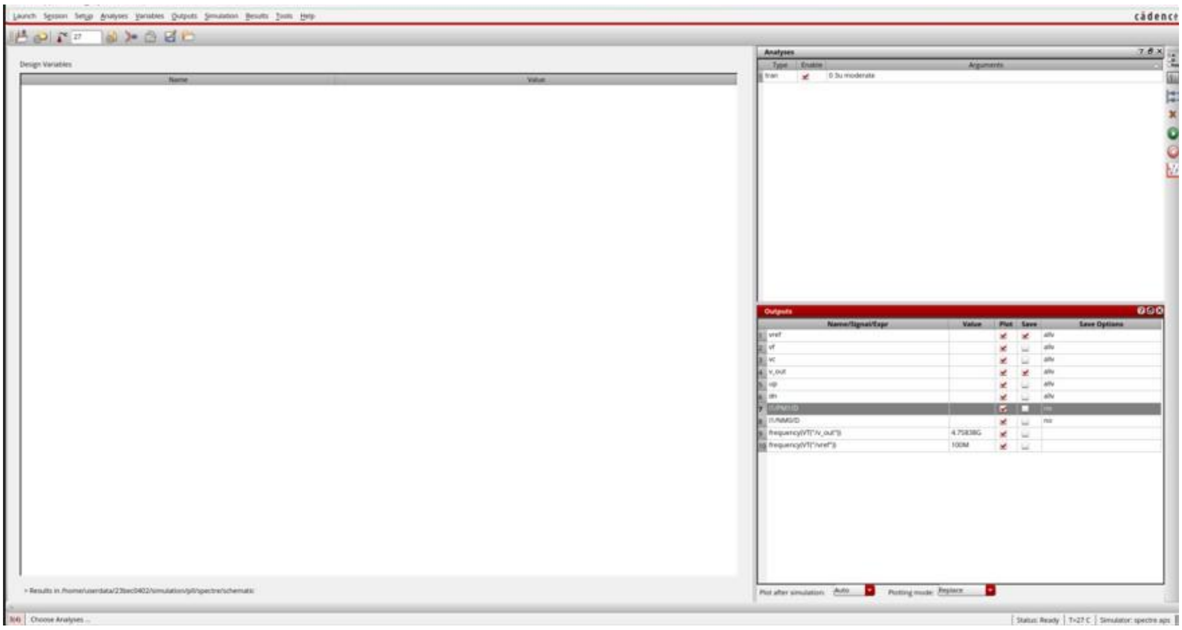
DIVIDER 48:

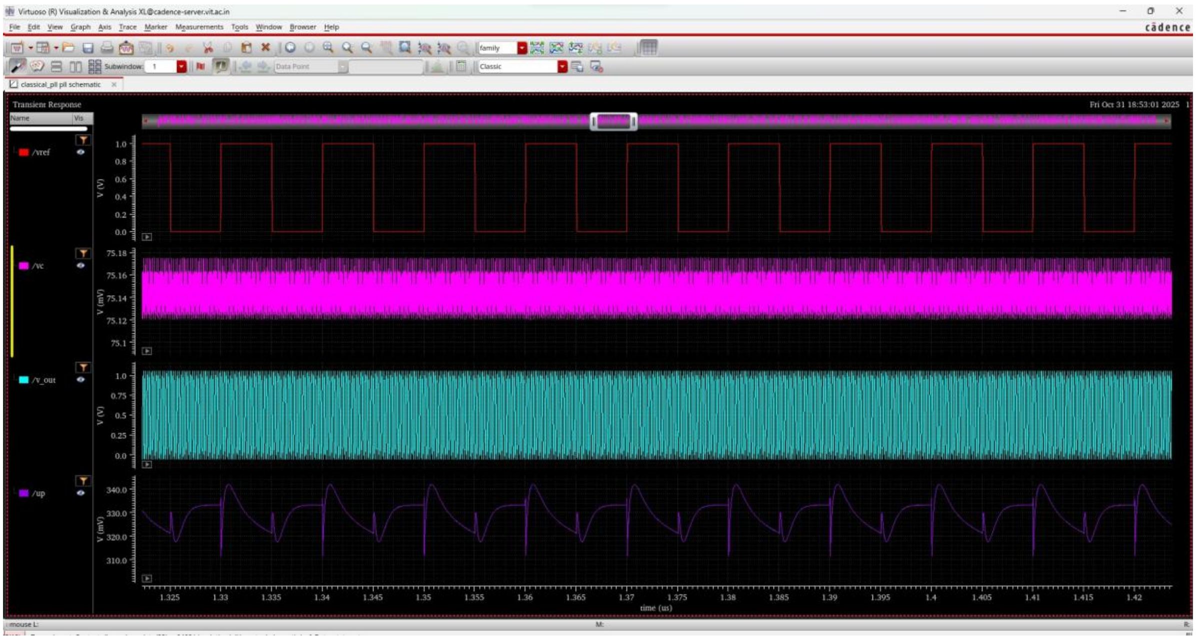


FINAL CIRCUIT:





OUTPUT:



Virtuoso (R) Visualization & Analysis XL@cadence-server.vit.ac.in
Transient Response
/vref
/vc
/V_out
/up
time (us)

Conclusion:

The PLL was designed and simulated using the 45 nm CMOS Technology. The supply voltage (VDD) was set to 1V. The Voltage-Controlled Oscillator (VCO) generates an oscillation frequency that varies approximately from ≈ 3.6 GHz to 4.9 GHz with respect to the control voltage (Vc), with a nominal operating point around ≈ 4.75 GHz for the locked condition

The PLL achieved a stable lock with the VCO output frequency multiplied with respect to the reference. In lock condition, the reference clock (Vref) and the divided VCO clock (Vdiv) are frequency- and phase-aligned. The control voltage (Vc) settles to a steady value indicating phase lock. The PLL lock time is observed to be approximately ≈ 120–150 µs, while the settling time (±5%) is around ≈ 35–45 µs.